

Canonical Correlation Analysis — Summary Sheet

What is CCA?

We have two groups of variables, $\mathbf{X}^{(1)}$ ($p \times 1$) and $\mathbf{X}^{(2)}$ ($q \times 1$). CCA finds linear combinations of each group — called **canonical variates** — that are maximally correlated with each other.

$$U_k = \mathbf{a}'_k \mathbf{X}^{(1)}, \quad V_k = \mathbf{b}'_k \mathbf{X}^{(2)}$$

The pairs (U_k, V_k) are found sequentially, each uncorrelated with all previous pairs. The correlation between U_k and V_k is the k -th **canonical correlation** ρ_k^* .

The maximum number of pairs is $\min(p, q)$, but in practice we only retain pairs with a meaningfully large ρ_k^* .

Partitioning the Covariance Matrix

The full covariance matrix is partitioned according to the two groups:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

where Σ_{11} is the within-group covariance of $\mathbf{X}^{(1)}$, Σ_{22} is for $\mathbf{X}^{(2)}$, and $\Sigma_{12} = \Sigma'_{21}$ captures between-group covariances. If working with standardised variables, Σ is replaced by the correlation matrix ρ .

For $U = \mathbf{a}' \mathbf{X}^{(1)}$ and $V = \mathbf{b}' \mathbf{X}^{(2)}$:

$$\text{Var}(U) = \mathbf{a}' \Sigma_{11} \mathbf{a}, \quad \text{Var}(V) = \mathbf{b}' \Sigma_{22} \mathbf{b}, \quad \text{Cov}(U, V) = \mathbf{a}' \Sigma_{12} \mathbf{b}$$

We maximise:

$$\text{Corr}(U, V) = \frac{\mathbf{a}'\Sigma_{12}\mathbf{b}}{\sqrt{\mathbf{a}'\Sigma_{11}\mathbf{a} \cdot \mathbf{b}'\Sigma_{22}\mathbf{b}}}$$

The Key Matrix $\mathbf{R}$ and Why $\Sigma_{11}^{-1/2}$ Appears

To solve the maximisation problem we make the substitution $\mathbf{a}^* = \Sigma_{11}^{1/2}\mathbf{a}$, so $\mathbf{a} = \Sigma_{11}^{-1/2}\mathbf{a}^*$. This turns the unit variance constraint $\mathbf{a}'\Sigma_{11}\mathbf{a} = 1$ into the simpler $\mathbf{a}^{*\prime}\mathbf{a}^* = 1$. The problem then reduces to a standard eigenvalue problem in $\mathbf{a}^*$, with solution given by the eigenvectors of:

$$\mathbf{R} = \Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11}^{-1/2}$$

To get back the canonical coefficients in the original variable space we undo the substitution:

$$\mathbf{a}_k = \Sigma_{11}^{-1/2}\mathbf{e}_k$$

where $\mathbf{e}_k$ is the k -th eigenvector of $\mathbf{R}$. So the $\Sigma_{11}^{-1/2}$ rescales the eigenvector back into the original variable space, accounting for the variance structure of $\mathbf{X}^{(1)}$.

The squared canonical correlations $\rho_1^{*2} \geq \rho_2^{*2} \geq \dots$ are the eigenvalues of $\mathbf{R}$.

Properties of Canonical Variates

$$\text{Var}(U_k) = \text{Var}(V_k) = 1$$

$$\text{Cov}(U_k, U_l) = \text{Cov}(V_k, V_l) = 0 \quad (k \neq l)$$

$$\text{Cov}(U_k, V_l) = 0 \quad (k \neq l)$$

$$\text{Corr}(U_k, V_k) = \rho_k^* \quad (\text{this is what we maximise})$$

So within each group, different canonical variates are uncorrelated. Across groups, only matched pairs (U_k, V_k) are correlated.

Canonical Loadings and Cross-Loadings

The **canonical coefficients** $\mathbf{a}_k$ and $\mathbf{b}_k$ are the weights on the original variables and should be avoided for interpretation due to multicollinearity.

Instead, use:

Canonical loadings — correlation between each original variable and its own group's canonical variate:

$$\text{Corr}(X_j^{(1)}, U_k), \quad \text{Corr}(X_j^{(2)}, V_k)$$

These tell you how much information U_k (or V_k) captures about each original variable.

Cross-loadings — correlation between original variables and the *other* group's canonical variate:

$$\text{Corr}(X_j^{(1)}, V_k), \quad \text{Corr}(X_j^{(2)}, U_k)$$

These exist because U_k and V_k are maximally correlated, so information from one group bleeds into the other's variate. The stronger ρ_k^* , the more informative the cross-loadings.

Signs matter — a positive loading means as the canonical variate increases, so does the original variable; negative means the opposite. If U_1 has positive loadings on $X_1^{(1)}$ and $X_2^{(1)}$ but negative cross-loadings on $X_1^{(2)}$ and $X_2^{(2)}$, this tells you the two groups move in opposite directions.

R Code: CCA from First Principles (Correlation Matrix)

```
# Example correlation matrix (p = q = 2)
rho <- matrix(c(1.0, 0.4, 0.5, 0.6,
               0.4, 1.0, 0.3, 0.4,
               0.5, 0.3, 1.0, 0.2,
               0.6, 0.4, 0.2, 1.0), 4, 4, byrow = TRUE)

# Step 1: Partition
p <- 2; q <- 2
```

```

rho11 <- rho[1:p, 1:p]
rho12 <- rho[1:p, (p+1):(p+q)]
rho21 <- rho[(p+1):(p+q), 1:p]
rho22 <- rho[(p+1):(p+q), (p+1):(p+q)]

# Step 2: Compute rho11^{-1/2} via eigendecomposition
# We need the matrix square root of rho11^{-1}
# Achieved by: V %%% diag(sqrt(eigenvalues)) %%% V^{-1}
rho11inv <- solve(rho11)
eig11 <- eigen(rho11inv)
rho11inv.sqrt <- eig11$vectors %%%
               diag(sqrt(eig11$values)) %%%
               solve(eig11$vectors)

# Step 3: Compute rho22^{-1}
rho22inv <- solve(rho22)

# Step 4: Form R and obtain eigendecomposition
R <- rho11inv.sqrt %%% rho12 %%% rho22inv %%% rho21 %%% rho11inv.sqrt
R_eig <- eigen(R)

# Step 5: Canonical correlations (square root of eigenvalues)
rho_star <- sqrt(R_eig$values)
rho_star

```

```
[1] 0.73872731 0.03014884
```

```

# Step 6: Canonical coefficients for U (a_k = rho11^{-1/2} e_k)
# Sign flip optional - just for alignment with cc() output
A <- -1 * rho11inv.sqrt %%% R_eig$vectors
A # columns are a_1, a_2, ...

```

```

      [,1]      [,2]
[1,] 0.8559647 -0.6766097
[2,] 0.2777371  1.0551485

```

```

# Step 7: Unstandardised canonical coefficients for V
# b_k is proportional to rho22^{-1} rho21 a_k
B_p <- rho22inv %%% rho21 %%% A

```

```
# Step 8: Standardise B so that  $\text{Var}(V_k) = \mathbf{b}_k' \boldsymbol{\rho}_{22} \mathbf{b}_k = 1$ 
# We need to divide each column of B_p by  $\sqrt{\mathbf{b}_k' \boldsymbol{\rho}_{22} \mathbf{b}_k}$ 
# This ensures the canonical variates  $V_k$  have unit variance
brhob <- sqrt(diag(t(B_p) %*% rho22 %*% B_p))
B <- B_p / matrix(brhob, nrow = nrow(B_p), ncol = ncol(B_p), byrow = TRUE)
B # columns are b_1, b_2, ...
```

```
      [,1]      [,2]
[1,] 0.5448119 -0.8630450
[2,] 0.7366455  0.7064135
```

The standardisation in Step 8 is necessary because $\mathbf{b}_k \propto \rho_{22}^{-1} \rho_{21} \mathbf{a}_k$ only guarantees the correct direction, not unit variance. We scale each column by $1/\sqrt{\mathbf{b}_k' \rho_{22} \mathbf{b}_k}$ so that $\text{Var}(V_k) = \mathbf{b}_k' \rho_{22} \mathbf{b}_k = 1$.

```
# Step 9: Canonical loadings
#  $\text{Corr}(X^{(1)}, U_k) = \mathbf{A} \boldsymbol{\rho}_{11}^{-1} \boldsymbol{\rho}_{12} \mathbf{b}_k$ 
loadings_U <- t(A) %*% rho11
loadings_U
```

```
      [,1]      [,2]
[1,] 0.9670595 0.6201230
[2,] -0.2545503 0.7845046
```

```
#  $\text{Corr}(X^{(2)}, V_k) = \mathbf{B} \boldsymbol{\rho}_{22}^{-1} \boldsymbol{\rho}_{21} \mathbf{a}_k$ 
loadings_V <- t(B) %*% rho22
loadings_V
```

```
      [,1]      [,2]
[1,] 0.6921410 0.8456079
[2,] -0.7217623 0.5338045
```

```
# Step 10: Cross-loadings
#  $\text{Corr}(X^{(1)}, V_k) = \mathbf{A} \boldsymbol{\rho}_{11}^{-1} \boldsymbol{\rho}_{12} \mathbf{b}_k$ 
cross_UV <- t(A) %*% rho12
cross_UV
```

```

      [,1]      [,2]
[1,] 0.51130348 0.62467366
[2,] -0.02176029 0.01609359

```

```

# Corr(X^(2), U_k) = B %%% rho21
cross_VU <- t(B) %%% rho21
cross_VU

```

```

      [,1]      [,2]
[1,] 0.714393288 0.4581018
[2,] -0.007674395 0.0236519

```

R Code: CCA Using the CCA Package

```

library(CCA)

# cc() takes the raw data matrices for each group
cc1 <- cc(X1, X2)

# Canonical correlations
cc1$cor

# Raw canonical coefficients
cc1$xcoef # a_k coefficients
cc1$ycoef # b_k coefficients

# Standardised coefficients (multiply by diagonal of SDs)
s1 <- diag(sqrt(diag(cov(X1))))
s2 <- diag(sqrt(diag(cov(X2))))
s1 %%% cc1$xcoef
s2 %%% cc1$ycoef

# Canonical variate scores for each observation
cc1$scores$xscores # U_k scores
cc1$scores$yscores # V_k scores

# Canonical loadings
cc1$scores$corr.X.xscores # Corr(X^(1), U_k)

```

```

cc1$scores$corr.Y.yscores # Corr( $X^{(2)}$ ,  $V_k$ )

# Cross-loadings
cc1$scores$corr.X.yscores # Corr( $X^{(1)}$ ,  $V_k$ )
cc1$scores$corr.Y.xscores # Corr( $X^{(2)}$ ,  $U_k$ )

```

Interpretation Summary

Quantity	What it tells you
ρ_k^*	How correlated the k -th pair of canonical variates are
$\mathbf{a}_k, \mathbf{b}_k$	Weights forming U_k and V_k — avoid for interpretation due to multicollinearity
$\text{Corr}(X_j^{(1)}, U_k)$	How much information U_k captures about $X_j^{(1)}$
$\text{Corr}(X_j^{(2)}, V_k)$	How much information V_k captures about $X_j^{(2)}$
$\text{Corr}(X_j^{(1)}, V_k)$	Cross-loading: how much $X_j^{(1)}$ bleeds into V_k via the canonical correlation
Sign of loading	Positive = move together; Negative = move in opposite directions